Composition Pair B1.15 + B1.16 — Action-Feedback Evolution as the Operational Mechanism for Bidirectional Evolution's Upward Vertical Direction, and Bidirectional Evolution as the Cross-Level Context That Makes Action-Feedback Evidence Architecturally Valuable Beyond Individual Entity Scope

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Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the enabling relationship between two architectural commitments from the source paper — action-feedback evolution (B1.15) and bidirectional evolution (B1.16) — as a composition pair in which each commitment depends on the other for a specific architectural property it cannot provide itself.

Abstract

The Coordination Knowledge Substrate (CKS) architecture for AI Selves commits both to action-feedback evolution (B1.15) — a governed pathway by which cell-level operational evidence informs DNA changes through proposal-and-approval cycles — and to bidirectional evolution (B1.16) — a two-axis model in which evolution proceeds horizontally within structural levels and vertically across them, with an upward direction from lower levels to higher and a downward direction from higher levels to lower. This note formalizes the composition pair formed by these two commitments, with particular attention to the upward vertical direction.

The enabling relationship runs as follows. B1.16 asserts that the upward vertical direction of evolution is architecturally available — that cell-level and aspect-level operational experience can inform evolution at aspect and Self scope respectively — but does not itself specify the mechanism by which this upward propagation is accomplished. B1.15's action-feedback pathway is that mechanism: the same proposal-and-approval cycle that governs DNA changes at cell scope is the pathway through which cell-level Action layer evidence reaches higher-level governed evolution decisions. Without B1.15, B1.16's upward direction is an architectural aspiration without an evidence-based governance pathway. Without B1.16's cross-level context, B1.15's evidence stays confined to the entity whose Action layer is

being examined, foreclosing the architecturally richer role in which cell-level operational experience informs the evolution of the aspects and Selves that compose those cells.

A secondary enabling relationship holds for B1.16's horizontal direction: when action-feedback evidence justifies a DNA change in one cell, the same evidence may identify peer cells with similar content domains as candidates for the same improvement, making B1.15 the evidence mechanism for horizontal evolution as well. The note covers pair identification, enabling direction, motivating direction, mutual dependency, co-presence analysis, the horizontal extension, and operational integration including detection signals and failure modes.

1. Why this composition pair requires explicit formalization

B1.15 and B1.16 are separately defensible commitments that appear in adjacent sections of the source paper (§7.2 and §7.4). Each can be read independently: action-feedback evolution is the mechanism that closes the loop between accumulated operational records and DNA changes; bidirectional evolution is the claim that evolution operates on two orthogonal axes, horizontal and vertical, with the vertical axis subdivided into upward and downward directions. A reader of each commitment in isolation would find them coherent without reference to the other.

The composition pair formalization is needed precisely because the separate readings obscure a structural dependency. B1.16's vertical-upward direction describes an architectural property — that lower-level operational experience can inform higher-level architecture evolution — without naming the mechanism through which that property is realized operationally. The mechanism is B1.15. This is not a minor coordination between two independently-specified features. The upward vertical direction of bidirectional evolution is only operationally meaningful if B1.15's proposal-and-approval pathway exists to carry cell-level evidence to higher-level governance decisions. A system that claims B1.16 but lacks B1.15 has the architectural aspiration of upward vertical evolution but no governed evidence-based pathway through which it is realized; it can evolve upward only through direct human judgment operating without the evidence advantage that operational records supply.

Conversely, B1.15 without B1.16 is a functioning mechanism whose evidence evaluations are scoped to the entity under examination. The mechanism works, but works narrowly: evidence in one cell's Action layer is evaluated for that cell's improvement, and the question of what that evidence implies for the aspect coordinating the cell, or the Self integrating the aspect, is not asked. B1.16's cross-level scope is what makes that question architecturally available and architecturally answerable. The composition pair formalization names this joint structure explicitly, so that implementations can verify whether both commitments are present and fully integrated rather than each present independently.

2. The enabling direction: B1.15 as the evidence mechanism for B1.16's upward vertical direction

Bidirectional evolution per B1.16 has a vertical dimension, and the vertical dimension has two directions (§7.4 of the source paper). The downward direction — higher-level architecture decisions propagating to lower levels — is addressed in the companion pair B4.09. This note addresses the upward direction: the movement of operational insight from lower levels (cells, aspects) to higher levels (aspects, Selves), where it informs architecture evolution at those higher levels.

B1.16 does not itself specify how upward vertical evolution is operationally realized. The source paper's formulation is that both axes operate through the three mechanisms (§7.4), and that the structural arrangements governing composition are themselves substrate content under human governance, making vertical evolution a substrate-edit operation. This is the architectural property; it says that the structure is editable in principle. What is needed in addition is the pathway through which lower-level operational experience — the lived record of what a cell does, how efficiently it does it, and where its orchestration falls short — reaches the governance decisions that would actually change higher-level architecture.

That pathway is B1.15's action-feedback evolution. The Action layer per B2.26 accumulates operational records: task instances, outputs, execution patterns, efficiency signals. Proposing substrates per B2.74 evaluate this evidence and identify improvement opportunities. The crucial architectural point for the B1.15 + B1.16 enabling pair is that proposing substrates do not evaluate only the question "should this cell's DNA change?" They also evaluate whether the evidence reveals that the aspect coordinating this cell should be architecturally different, or whether the Self integrating this aspect should evolve. When cell-level operational experience indicates that a coordination arrangement at aspect scope is inefficient, or that an integration architecture at Self scope produces systematic friction, that is the upward vertical signal per B2.81. B1.15's two-stage governance per B2.75 — Stage 1 review and Stage 2 approval — is how that signal becomes a governed DNA change at the appropriate level: cell DNA through cell-scope directed selection, aspect DNA through aspect-scope directed selection, Self DNA through Self-scope directed selection.

The enabling direction is therefore precise: B1.15 provides the evidence accumulation (Action layer per B2.26), the evidence evaluation with cross-level scope (proposing substrates per B2.74), and the governed approval pathway (two-stage governance per B2.75) that together operationalize B1.16's upward vertical direction. Without B1.15, the upward vertical direction of B1.16 has no evidence mechanism. Governance could still make upward vertical changes through directed selection based on human judgment, but the specific evidence advantage — that operational records from the Action layer inform the change rather than human judgment alone — would be absent.

3. The motivating direction: B1.16 as the cross-level scope for B1.15

The relationship runs in both directions. B1.15 enables B1.16's upward direction; B1.16 provides the motivating context that makes B1.15's evidence architecturally valuable beyond the scope of the individual entity being examined.

Without B1.16, action-feedback evolution improves individual entities. Evidence in a cell's Action layer is evaluated for that cell's DNA. If the evaluation reveals a systematic inefficiency, that cell's DNA changes. This is useful and is the core commitment of B1.15. But it is architecturally bounded: the evidence loop closes at cell scope, and the same evidence is never asked what it implies at aspect or Self scope.

With B1.16's cross-level scope, the evaluation context is richer. The same evidence that justifies a cell-level DNA change is also evaluated for cross-level implications: does this evidence pattern, observed across multiple cells within an aspect, indicate that the aspect's coordination rules should change? Does this evidence, accumulated over many operational cycles, suggest that the Self's integration architecture should be restructured? B1.16 provides the architectural rationale for asking these questions — the upward vertical direction is defined as a legitimate evolution pathway, so evidence at lower levels is legitimately evaluable for higher-level implications — and B1.15's governance pathway provides the mechanism for acting on the answers.

The result is that B1.15's evidence becomes a multi-level improvement mechanism in the presence of B1.16, not a single-level one. The action-feedback loop does not just close between the Action layer and the DNA layer of a single cell; it closes between cell-level operational experience and the entire architectural stack that cell's operations are embedded in. This is the motivating direction: B1.16 does not mechanically enable B1.15 (B1.15 functions without it, just more narrowly), but B1.16 provides the cross-level context that gives B1.15's evidence its full architectural value.

4. Mutual dependency and the co-presence requirement

The mutual dependency is asymmetric in one important respect: B1.15 is a functioning mechanism without B1.16, while B1.16's upward vertical direction is an architectural aspiration without an evidence mechanism in the absence of B1.15. The dependency is therefore stronger in one direction — B1.16 depends on B1.15 for operational realization of its upward direction — than in the other, where B1.16 provides scope amplification rather than basic operationalization.

This asymmetry is architecturally significant. It means that a system implementing only B1.15 has a working action-feedback mechanism but a narrowly scoped one; implementing only B1.16 claims bidirectional evolution but lacks the evidence-based pathway for its upward direction. Neither absence is total failure of functionality — directed selection per B1.14 can still produce upward vertical changes through human judgment alone, and B1.15 still produces cell-level improvements — but both absences represent the loss of an architectural property the source paper commits to when both B1.15 and B1.16 are present.

B1.15 without B1.16: Action-feedback improves individual cells. Evidence is never evaluated for cross-level implications. Proposing substrates evaluate cell-scope improvement only. Cell improvements remain at cell scope. The architectural question — does this cell-level evidence imply that the aspect should be restructured? — is never asked, because the upward vertical direction of B1.16 is not present to make the question architecturally motivated.

B1.16 without B1.15: Bidirectional evolution is present as architectural commitment, including the upward vertical direction. But the upward direction lacks an evidence mechanism. Upward vertical evolution occurs only through directed selection based on human judgment, without the operational evidence advantage B1.15 provides. The action-feedback loop does not supply evidence for higher-level governance decisions; those decisions must be made on other grounds.

Full co-presence: Cell-level operational evidence is evaluated for both cell-level and cross-level improvement opportunities. Proposing substrates identify evidence patterns that support changes at cell, aspect, or Self scope. Two-stage governance authorizes changes at the appropriate scope. Upward vertical evolution is evidence-based and governed. The architectural promise of B1.16's upward direction is operationally realized through B1.15's pathway.

5. Horizontal evolution as shared beneficiary

The B1.15 + B1.16 enabling relationship is not limited to the vertical axis. B1.16's horizontal direction — evolution that refines content within existing structure, where existing cells refine their DNA and existing aspects refine their cell composition — also uses B1.15 as its evidence mechanism (§7.4 of the source paper).

When one cell's Action layer evidence identifies a DNA improvement, that same evidence may carry implications for peer cells with similar content domains within the same aspect. If cell A's operational records reveal that a particular orchestration pattern is inefficient for its content domain, and cells B and C handle the same content domain under structurally similar orchestration, then the evidence from cell A's Action layer is evidence for cells B and C's DNA as well. B1.15's proposing substrates can identify this peer-relevance: Stage 1 review and Stage 2 approval authorize DNA changes for peer cells when the evidence supports them. This is horizontal evolution realized through the action-feedback pathway.

The architectural significance is that B1.16's two directions — horizontal and vertical-upward — both draw on B1.15 as their evidence mechanism, making B1.15 the operational foundation for bidirectional evidence-based evolution in both the structural and cross-level axes. This is why the presence of both commitments produces substantially richer evolution dynamics than either alone: B1.15 supplies the evidence and governance pathway, B1.16 supplies the multi-directional architectural scope through which that evidence is evaluated and applied.

6. Operational integration, detection, and failure modes

Operational integration. The cell-level Action layer per B2.26 accumulates operational records through cell execution. Proposing substrates per B2.74 evaluate these records for improvement opportunities, with evaluation scope extended by B1.16's cross-level context to include aspect-level and Self-level implications. Stage 1 governance per B2.75 reviews proposals including cross-level implications identified by the proposing substrates. Stage 2 approval per B2.75 authorizes DNA changes at the appropriate level: cell DNA through cell-scope directed selection, aspect DNA through aspect-scope directed selection, Self DNA through Self-scope directed selection. B1.16's upward vertical direction is realized at the moment Stage 2 approval authorizes a higher-level DNA change based on lower-level evidence — the Action layer at cell scope has informed a governed change at aspect or Self scope.

B1.16’s horizontal direction is realized when Stage 2 approval authorizes peer-cell DNA changes based on evidence from one cell’s Action layer.

Detection. Two verification references from the source paper are relevant. B2.78 covers action-feedback verification and specifically addresses pathway integrity: the diagnostic question is whether the complete pathway from evidence to higher-level DNA change has ever been executed, not merely whether the pathway is defined. A system in which no proposal has ever crossed a scope boundary — where all approved proposals target only the cell whose Action layer generated the evidence — has not operationally realized the B1.15 + B1.16 pair even if both commitments are nominally present. B2.83 covers bidirectional evolution verification and specifically addresses whether the upward vertical evolution pathway is operational. The pair-completeness detection signal is whether action-feedback proposals ever address higher-level DNA. If the answer is consistently no, the co-presence is nominal rather than operational.

Failure modes. Evidence Blindness (B3.28) names the anti-pattern in which B1.15’s evidence pathway is broken — operational records in the Action layer do not reach proposing substrates, or proposing substrates do not produce proposals, or the governance cycle does not complete. When Evidence Blindness is present, B1.16’s upward direction loses its evidence mechanism; upward vertical evolution is not impossible, but it is reduced to directed selection based on human judgment alone. Unidirectional Evolution (B3.17) names the anti-pattern in which B1.16’s upward direction is absent — evolution operates only horizontally or only downward, without the upward propagation of lower-level operational experience to higher-level architecture decisions. When Unidirectional Evolution is present, B1.15’s evidence evaluations are confined to cell scope regardless of what the evidence implies about higher-level architecture. Both anti-patterns can be present independently or together; their co-occurrence produces a system in which action-feedback at cell scope neither proposes nor receives cross-level implications.

Source paper

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